

ASSIGNMENT 1: DATA PRE-PROCESSING

Completion Requirements and Technical Report Specification

Item	Details
Opened	Monday, 3 August 2026, 12:00 AM
Due	Friday, 21 August 2026, 5:30 PM
Report limit	Maximum four pages of written content, excluding the notebook
Audience	Head of Credit Risk at the lending institution
Focus	Quality, reliability, fairness, and reproducibility of the machine learning solution

Assignment Instructions

Extend your Lab 2 notebook into a concise technical report (maximum four pages of written content, excluding the notebook). The report should be written for the Head of Credit Risk at the lending institution, focusing on the quality, reliability, and reproducibility of your machine learning solution.

1. Data Quality Audit — 5 Marks

Conduct a comprehensive audit of the dataset and identify the ten columns with the most significant data quality issues (refer to Figure 2.1).

- The data quality issue identified.
- The evidence supporting your conclusion (e.g., summary statistics, missing value analysis, duplicate counts, diagnostic plots, or validation checks).
- The missing-data classification where applicable: MCAR, MAR, or MNAR.
- The remediation strategy selected and why it is appropriate.

Present the findings in a well-organised table.

Column	Issue	Evidence	MCAR/MAR/MNAR	Remediation & rationale
Column 1	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 2	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 3	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 4	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 5	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 6	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 7	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 8	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 9	Identify issue	Provide evidence	If applicable	State treatment and justification
Column 10	Identify issue	Provide evidence	If applicable	State treatment and justification

2. Feature Dictionary — 5 Marks

Create and document at least 15 engineered features drawn from a minimum of four feature families listed in Table 2.2.

- Feature name.
- Definition: what the feature represents.
- Computation window: the time period over which it is calculated.
- Upstream fields: the data lineage used to construct the feature.
- Write time of every upstream field: when each source field becomes available.

- A brief fairness note, where relevant (see Remark 2.3), describing any potential bias or ethical consideration associated with the feature.

Feature	Definition	Window	Upstream fields	Write time	Fairness note
Feature 1	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 2	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 3	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 4	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 5	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 6	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 7	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 8	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 9	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 10	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 11	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 12	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 13	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 14	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration
Feature 15	Define meaning	e.g. 30 days	Source fields	Availability time	Bias/ethics consideration

3. Leakage Assessment Memo — 6 Marks

Prepare a short technical memo describing the two intentionally planted sources of data leakage.

- Identify where each leakage source occurs.
- Explain how the leakage happens by referring to the six causes of data leakage discussed in Section 2.7.2.
- Quantify the impact by comparing model performance before and after the leakage is removed, for example the change in offline ROC-AUC.
- Recommend a practical process or governance change that would prevent this type of leakage in future model development.

Leakage source	Where it occurs	Cause	Impact: before vs after	Governance prevention
Leakage 1	Identify location	Map to Section 2.7.2	ROC-AUC before → after	Recommended control
Leakage 2	Identify location	Map to Section 2.7.2	ROC-AUC before → after	Recommended control

4. Reproducible Machine Learning Pipeline — 4 Marks

Develop a single, end-to-end Scikit-learn Pipeline that performs all preprocessing and model training.

- Perform all preprocessing steps within the pipeline.
- Evaluate the model using group-aware cross-validation.
- Report the final cross-validated ROC-AUC.
- Save the trained pipeline using joblib.
- Ensure the saved pipeline can be reloaded and executed successfully by the marker without modification.

Important Requirements

- Every conclusion or design decision must be supported with evidence, not assumptions.
- Any preprocessing performed outside the cross-validation pipeline will be treated as data leakage and will result in the loss of the pipeline marks, regardless of the model's performance.
- The emphasis of this assignment is on producing a solution that is correct, reproducible, well-documented, and suitable for deployment, rather than simply achieving the highest predictive accuracy.

Submission Checklist

✓	Report is no more than four pages of written content, excluding the notebook.
✓	Ten most significant data-quality problems are supported by evidence.
✓	Missingness is classified as MCAR, MAR, or MNAR where applicable.
✓	At least 15 engineered features come from at least four feature families.
✓	Feature lineage and source-field write times are documented.
✓	Fairness considerations are documented where relevant.
✓	Both planted leakage sources are identified and their impact quantified.
✓	A single end-to-end Scikit-learn Pipeline contains all preprocessing and training.
✓	Group-aware cross-validation is used and ROC-AUC is reported.
✓	The final pipeline is saved with joblib and successfully reloadable.